

CV-Mamba: Gated CNN Transfer Learning for Mango Leaf Disease Detection with pHash-Based Deduplication and Leakage-Free Partitioning

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2 ABSTRACT

Mango leaf disease poses a significant agricultural threat in tropical regions, necessitating robust and scalable automated diagnostic systems. This study presents, to the best of our knowledge, the first comprehensive deep learning pipeline for eight-class mango leaf disease classification. The pipeline is enhanced with perceptual hash-based deduplication and stratified, leakage-free data partitioning to ensure reliable benchmarking across five MambaOut variants: Femto, Kobe, Tiny, Small, and Base, under a unified transfer learning protocol. A perceptual hashing (pHash) approach with a Hamming distance threshold of $\tau = 8$ was employed to identify and remove near-duplicate samples. This process eliminated 739 images (18.48%) from the original dataset of 4,000 images, resulting in 3,261 unique samples distributed across eight disease classes. To ensure data integrity, a five-step protocol was implemented, including pre-split deduplication, stratified shuffle splitting (70%:15%:15%), strict isolation of data transformations, and normalization using ImageNet statistics. This design guarantees statistical independence across training, validation, and test sets. An extensive stochastic data augmentation strategy was

applied exclusively to the training set to prevent distribution leakage. The augmentation pipeline included random resized cropping, horizontal and vertical flipping, affine transformations with variable shifts, color jittering (brightness and contrast), Gaussian blur, and random erasing. All MambaOut backbones were initialized with ImageNet-1K pretrained weights and kept frozen during training. Only a custom three-layer classification head was trained using the Adamax optimizer, label-smoothed cross-entropy loss ($\epsilon = 0.1$), and a cosine annealing learning rate scheduler over 10 epochs. Experimental results demonstrate that MambaOut-Small achieves the best performance, with a test accuracy of 99.59%, a macro F1-score of 0.9962 (90% confidence), and a weighted F1-score of 0.9959. Notably, it outperforms the larger MambaOut-Base model (99.39% accuracy), indicating a non-monotonic relationship between model size and transfer learning performance. For resource-constrained or edge deployment scenarios, MambaOut-Kobe (9.1M parameters) offers an excellent trade-off, achieving 99.18% accuracy with approximately $5.3\times$ fewer parameters than the top-performing model.

Keywords: keyword, keyword, keyword, keyword, keyword, keyword, keyword, keyword

1 INTRODUCTION

Mango (*Mangifera indica* L.) is one of the most economically significant tropical fruit crops worldwide, contributing substantially to agricultural output and food security across South and Southeast Asia, Africa, and Latin America Singh and Bhargava (2020). Despite its agronomic importance, mango cultivation is severely threatened by a diverse spectrum of foliar diseases — including *Anthraxnose*, *Bacterial Canker*, *Powdery Mildew*, *Sooty Mould*, *Gall Midge*, *Die Back*, and *Cutting Weevil* — which collectively cause yield losses estimated between 20% and 80% if left undiagnosed in early stages Kumar et al. (2021). Traditional disease diagnosis relies on visual inspection by trained agronomists, a process that is inherently subjective, labor-intensive, geographically constrained, and impractical at the scale demanded by modern precision agriculture.

The rapid advancement of deep learning, particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), has fundamentally transformed the landscape of plant disease recognition, enabling automated, scalable, and objective diagnosis from digital imagery LeCun et al. (2015); Dosovitskiy et al. (2020). Transfer learning, in which models pretrained on large-scale datasets such as ImageNet-1K are fine-tuned for downstream agricultural tasks, has emerged as the dominant paradigm for achieving strong performance under limited labelled data conditions Pan and Yang (2009). However, the majority of existing plant disease classification studies suffer from methodological limitations that undermine the validity and reproducibility of their reported results — most critically, inadequate deduplication of training data, failure to verify disjointness of train/test partitions, and application of augmentation transforms to evaluation sets, all of which constitute forms of data leakage Kapoor and Narayanan (2023).

The recent introduction of **MambaOut** Yu and Wang (2025), a family of purely Gated CNN-based vision architectures operating within the MetaFormer paradigm, presents a compelling alternative to both CNN and Transformer backbones. By demonstrating that selective state space model (SSM) components — central to the original Mamba architecture — are unnecessary for image-level recognition tasks, MambaOut achieves competitive ImageNet-1K accuracy with substantially reduced architectural complexity. Despite its strong performance on general vision benchmarks, the applicability of MambaOut to fine-grained agricultural image classification tasks such as mango disease recognition has not been systematically investigated.

57 This paper addresses these gaps through four primary contributions:

- 58 1. A rigorous pHash-based deduplication protocol that removes 739 near-duplicate images (18.48%) from
59 the raw dataset prior to any partitioning, preventing cross-split contamination.
- 60 2. A five-point data leakage prevention framework enforcing complete statistical independence across
61 training, validation, and test partitions.
- 62 3. A comprehensive benchmark of all five MambaOut variants — Femto, Kobe, Tiny, Small, and Base —
63 under a unified frozen-backbone transfer learning protocol with task-isolated augmentation.
- 64 4. A multi-metric evaluation comprising accuracy, macro/weighted F1, AUC-ROC, precision-recall
65 curves, confidence distribution analysis, and true/false prediction visualization.

66 Experimental results demonstrate that MambaOut-Small achieves state-of-the-art performance with
67 99.59% accuracy and a macro F1-score of 0.9962 on the eight-class Mango Leaf Disease Dataset, while
68 MambaOut-Kobe provides an efficient edge-deployable alternative at 99.18% accuracy with only 9.1M
69 parameters.

2 LITERATURE REVIEW

70 2.1 Plant Disease Detection using Deep Learning

71 The application of deep learning to automated plant disease detection has grown exponentially since
72 the seminal work of Mohanty et al. Mohanty et al. (2016), who demonstrated that a CNN trained on
73 the PlantVillage dataset could identify 26 diseases across 14 crop species with over 99% accuracy under
74 controlled laboratory conditions. Subsequent studies extended this work to field conditions, revealing a
75 substantial performance gap attributable to background complexity, illumination variability, and multi-
76 scale disease presentation Brahim et al. (2017). Ferentinos Ferentinos (2018) evaluated multiple CNN
77 architectures on a large-scale plant disease dataset and established AlexNet and VGG as strong baselines,
78 while Barbedo Barbedo (2018) systematically analysed the impact of image resolution, dataset size, and
79 class imbalance on classification performance.

80 The transition from traditional CNNs to residual architectures marked a significant inflection point in
81 the field. He et al. He et al. (2016) introduced ResNet, which enabled training of very deep networks
82 through skip connections, and subsequent works demonstrated its superior transferability to plant disease
83 recognition tasks Too et al. (2019). EfficientNet Tan and Le (2019), which scales network width, depth, and
84 resolution via a compound coefficient, became widely adopted as a lightweight yet accurate backbone for
85 agricultural image classification. Inception-based architectures Szegedy et al. (2016) and DenseNet Huang
86 et al. (2017) were similarly applied, with ensemble methods combining multiple backbone predictions
87 further boosting robustness Johannes et al. (2017).

88 2.2 Mango Leaf Disease Classification

89 Mango-specific disease recognition has attracted increasing research attention in recent years. Iqbal
90 et al. Iqbal et al. (2021) proposed a CNN-based system for mango leaf disease classification achieving
91 97.8% accuracy, utilising handcrafted feature extraction as a preprocessing stage. Mittal et al. Mittal et al.
92 (2022) investigated the application of MobileNetV2 and InceptionV3 for detecting five mango diseases,
93 reporting that transfer learning from ImageNet significantly outperformed training from scratch under the
94 constrained data conditions typical of agricultural field datasets. Prawira et al. Prawira et al. (2022) applied
95 data augmentation strategies including rotation and flipping to address class imbalance, observing improved

recall on minority disease classes. Ahmed et al. (2023) benchmarked seven pretrained CNN architectures on the publicly available Mango Leaf Disease Dataset, identifying ResNet50 and EfficientNetB3 as top performers, though the study did not apply deduplication prior to splitting, raising concerns about data integrity. Collectively, these studies demonstrate the viability of deep transfer learning for mango disease classification while highlighting the need for more rigorous experimental protocols.

2.3 Vision Transformers and Hybrid Architectures

The introduction of the Vision Transformer (ViT) by Dosovitskiy et al. (2020) established self-attention as a competitive alternative to convolution for visual recognition. ViT operates by splitting images into fixed-size patches, projecting them into embedding vectors, and processing them through standard Transformer encoder blocks, achieving strong performance when pretrained on large datasets such as JFT-300M. Subsequent hybrid architectures sought to combine the local inductive biases of CNNs with the global receptive field of attention mechanisms. DeiT (Touvron et al., 2021) enabled ViT training on ImageNet alone through knowledge distillation, while Swin Transformer (Liu et al., 2021) introduced shifted window attention to reduce computational complexity from quadratic to linear in image size. These architectures were subsequently adopted for agricultural image analysis, with several studies reporting that Swin Transformer outperforms ResNet variants on fine-grained crop disease tasks (Chen et al., 2022).

2.4 State Space Models and Mamba

Structured State Space Models (S4) (Gu et al., 2022) emerged as a compelling sequence modelling paradigm, capable of capturing long-range dependencies with linear computational complexity through efficient convolution-based computation in the frequency domain. Mamba (Gu and Dao, 2023) extended S4 with input-dependent selective state space transitions, enabling dynamic context-aware sequence processing and achieving competitive performance with Transformers on language modelling benchmarks. Vision Mamba (Vim) (Zhu et al., 2024) and VMamba (Liu et al., 2024) adapted the Mamba block for 2D visual data by introducing bidirectional scanning strategies over image patches, demonstrating competitive ImageNet accuracy while maintaining sub-quadratic complexity. MambaND (Li et al., 2024) further generalised the scanning mechanism to multi-dimensional inputs. These architectures attracted significant attention as potential successors to Transformers for visual recognition, particularly for high-resolution and long-context tasks.

2.5 MambaOut: Revisiting the Necessity of SSMs

Yu et al. (2025) challenged the assumption that SSM components are necessary for strong visual recognition performance by introducing **MambaOut** (CVPR 2025). Drawing on the MetaFormer framework (Yu et al., 2022), which posits that the general architecture of token mixing and channel mixing is more important than the specific mixing operator, they replaced the SSM-based token mixer with a simple Gated CNN block — a depthwise convolution gated by a GELU-activated linear projection. Despite this architectural simplification, MambaOut variants achieve competitive or superior ImageNet-1K accuracy compared to full Mamba vision models, supporting the hypothesis that the selective SSM mechanism provides limited benefit for image classification tasks where causal sequential modelling is unnecessary. The MambaOut family spans five scales — Femto (7.3M), Kobe (9.1M), Tiny (26.5M), Small (48.5M), and Base (84.8M) — enabling systematic evaluation of the parameter-performance trade-off. To the best of our knowledge, this work represents the first systematic application of all MambaOut variants to agricultural leaf disease classification.

2.6 Data Integrity in Deep Learning Benchmarks

A growing body of literature has identified data leakage as a pervasive source of inflated performance estimates in machine learning research. Kapoor and Narayanan (2023) conducted a systematic review of 294 papers across multiple domains and identified leakage in 67% of reviewed studies, attributing overestimation of accuracy by up to 30% in some cases. In the specific context of image classification, near-duplicate contamination — where visually similar or identical images appear in both training and test sets — has been documented in large-scale datasets including ImageNet (Deng et al., 2009). Zhu et al. (2022) demonstrated that perceptual hash-based deduplication using pHash significantly reduces cross-split leakage and produces more conservative, reliable generalisation estimates. Despite this evidence, the majority of plant disease classification studies do not report deduplication procedures, and fewer still verify partition disjointness programmatically, motivating the rigorous preprocessing protocol employed in this work.

3 METHODOLOGY

3.1 Dataset Description and Characteristics

This study employs the Mango Leaf Disease Dataset, a publicly available benchmark comprising 4,000 high-resolution RGB field images captured under natural lighting conditions across eight pathological and healthy categories: *Anthracnose*, *Bacterial Canker*, *Cutting Weevil*, *Die Back*, *Gall Midge*, *Healthy*, *Powdery Mildew*, and *Sooty Mould*. Images were acquired using handheld cameras with variable backgrounds, illumination intensities, and viewing angles, making the dataset representative of real-world agricultural deployment scenarios. Prior to any model development, a rigorous preprocessing pipeline encompassing perceptual deduplication, statistical exploratory analysis, and stratified partitioning is applied to ensure dataset integrity and experimental reproducibility.

3.2 Perceptual Hash-Based Deduplication

Near-duplicate images introduce artificial statistical correlations between partitions, constituting a form of implicit data leakage that can cause overestimated generalization metrics. To systematically detect and remove such redundancies, we employ **Perceptual Hashing (pHash)**, a Discrete Cosine Transform (DCT)-based image fingerprinting method that generates a compact binary representation of each image's low-frequency visual structure, invariant to minor resolution differences, JPEG compression artifacts, brightness shifts, and minor geometric perturbations.

The pHash computation proceeds through the following formal steps:

Step 1 — Spatial downsampling and luminance conversion:

$$I' = \text{Resize}(I, 32 \times 32), \quad I_g = 0.299 I_R + 0.587 I_G + 0.114 I_B \quad (1)$$

where I_R , I_G , I_B are the red, green, and blue channel matrices of the input image I , and $I_g \in \mathbb{R}^{32 \times 32}$ is the resulting luminance image.

Step 2 — 2D Discrete Cosine Transform (DCT-II):

$$F(u, v) = \frac{2}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} I_g(x, y) \cos \left[\frac{\pi u(2x+1)}{2N} \right] \cos \left[\frac{\pi v(2y+1)}{2N} \right] \quad (2)$$

where $N = 32$, and only the top-left 8×8 low-frequency submatrix is retained:

$$\hat{F} = F[0 : 8, 0 : 8] \in \mathbb{R}^{64} \quad (3)$$

Step 3 — Binary hash generation via mean thresholding:

$$\bar{F} = \frac{1}{64} \sum_{i=1}^{64} \hat{F}_i, \quad h_i = \begin{cases} 1 & \text{if } \hat{F}_i > \bar{F} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

yielding a 64-bit binary hash $\mathbf{h} \in \{0, 1\}^{64}$ per image.

Step 4 — Pairwise Hamming distance duplicate detection:

$$d_H(\mathbf{h}_a, \mathbf{h}_b) = \sum_{i=1}^{64} \mathbf{h}_a^{(i)} \oplus \mathbf{h}_b^{(i)} \leq \tau, \quad \tau = 8 \quad (5)$$

where $\oplus$ is the bitwise XOR operator. The threshold $\tau = 8$ corresponds to a maximum Hamming distance of 12.5% of the 64 hash bits. When a duplicate is detected, the later-indexed image is removed, preserving the first encountered instance to maintain reproducibility. Application of this procedure identified and removed $N_{\text{dup}} = 739$ near-duplicate images (18.48%), yielding a clean corpus of $N = 3,261$ unique images whose class distribution is summarised in Table 1.

Table 1. Class distribution after pHash deduplication.

Class	Count	Proportion (%)
Bacterial Canker	471	14.44
Powdery Mildew	459	14.07
Anthraco	458	14.04
Sooty Mould	447	13.70
Gall Midge	427	13.09
Die Back	388	11.90
Healthy	374	11.47
Cutting Weevil	237	7.27
Total	3,261	100.00

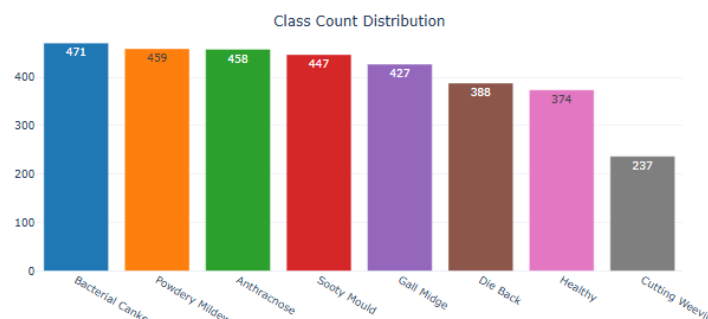


Figure 1. Class count distribution across eight mango leaf disease categories after pHash-based deduplication, showing the number of retained images per class from the original 4,000-image dataset ($N = 3,261$).

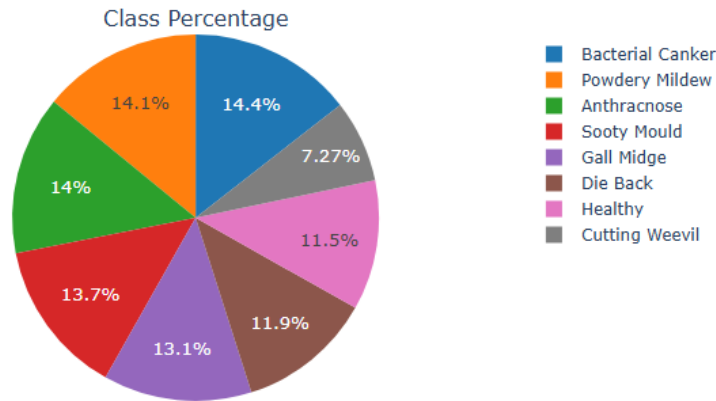


Figure 2. Proportional class distribution of the deduplicated Mango Leaf Disease Dataset across eight pathological and healthy categories, illustrating mild class imbalance with *Cutting Weevil* as the minority class (7.27%).

180 3.3 Exploratory Data Analysis

181 Prior to model training, a comprehensive Exploratory Data Analysis (EDA) is conducted to characterise
182 the visual properties of the dataset and identify any systematic biases or anomalies that may affect learning.

183 **Spatial dimension analysis** computes the width w_i and height h_i of each image and examines their joint
184 distribution:

$$\bar{w} = \frac{1}{N} \sum_{i=1}^N w_i, \quad \bar{h} = \frac{1}{N} \sum_{i=1}^N h_i, \quad \rho_i = \frac{w_i}{h_i} \quad (6)$$

185 **Channel-wise intensity analysis** quantifies the mean pixel intensity per RGB channel for each class c :

$$\mu_{c,k} = \frac{1}{|S_c| \cdot H \cdot W} \sum_{i \in S_c} \sum_{x=1}^H \sum_{y=1}^W I_i^{(k)}(x, y), \quad k \in \{R, G, B\} \quad (7)$$

186 where $S_c = \{i : y_i = c\}$ is the index set of class c . Differences in $\mu_{c,G}$ are particularly diagnostic for
187 distinguishing healthy foliage from diseased tissue.

188 **Grayscale pixel intensity histograms** are estimated per class:

$$I_{g,i} = \text{Grayscale}(\text{Resize}(I_i, 64 \times 64)), \quad \hat{p}_c(v) = \frac{1}{|S_c| \cdot 4096} \sum_{i \in S_c} \sum_{x,y} \mathbf{1}[I_{g,i}(x, y) = v] \quad (8)$$

189 for intensity bins $v \in \{0, 1, \dots, 255\}$, enabling visual assessment of inter-class separability at the pixel
190 level.

191 3.4 Stratified Data Partitioning and Leakage Prevention

192 To construct statistically valid and bias-free training, validation, and test partitions, we apply **Stratified**
193 **Shuffle Split (SSS)**, which enforces proportional class representation in each subset. Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$

194 denote the clean dataset with $C = 8$ classes. The dataset is partitioned into three strictly disjoint subsets:

$$\mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}} = \mathcal{D} \quad (9)$$

195

$$\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{val}} = \mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{test}} = \mathcal{D}_{\text{val}} \cap \mathcal{D}_{\text{test}} = \emptyset \quad (10)$$

196 with proportions 70%:15%:15%. The stratification constraint enforces that for every class c and split s :

$$\left| \frac{|\{i : y_i = c, x_i \in \mathcal{D}_s\}|}{|\mathcal{D}_s|} - \frac{|\{i : y_i = c\}|}{N} \right| \leq \delta \quad (11)$$

197 where δ is the small tolerance enforced by the stratification algorithm. Both split stages employ a fixed
198 random seed $s = 123$ for reproducibility.

199 The following data integrity constraints are programmatically enforced and verified via set-intersection
200 assertions before any model is instantiated:

- 201 1. **Spatial disjointness:** No file path appears in more than one split.
- 202 2. **Pre-split deduplication:** pHash removal is applied to the entire dataset before partitioning, ensuring
203 no near-duplicate spans two splits.
- 204 3. **Transform isolation:** Stochastic augmentation is bound exclusively to $\mathcal{D}_{\text{train}}$; evaluation sets receive
205 only deterministic preprocessing.
- 206 4. **Normalization source:** Mean and standard deviation statistics originate from ImageNet pretraining
207 configurations — never estimated from the training partition — eliminating normalization-induced
208 leakage.
- 209 5. **Loader integrity:** Validation and test `DataLoader` instances operate with `shuffle=False` and
210 zero stochastic transforms.

211 3.5 Data Augmentation

212 Augmentation is applied **exclusively** to $\mathcal{D}_{\text{train}}$ as a regularization mechanism to improve generalization.
213 For a given training image x , the full stochastic augmentation pipeline is:

$$\tilde{x} = \mathcal{T}_{\text{train}}(x) = \mathcal{N}(\mathcal{E}(\mathcal{B}(\mathcal{A}(\mathcal{G}(\mathcal{F}(x))))) \quad (12)$$

214 **Random Resized Crop $\mathcal{F}$:** A random crop of scale $s \sim \mathcal{U}(0.7, 1.0)$ and aspect ratio $r \sim \mathcal{U}(0.85, 1.15)$ is
215 extracted and resized to $H \times W$:

$$\mathcal{F}(x) = \text{Resize}(\text{Crop}(x, s, r), H \times W) \quad (13)$$

216 **Random Flips $\mathcal{G}$:** Horizontal reflection with $p_H = 0.5$ and vertical reflection with $p_V = 0.3$:

$$\mathcal{G}(x) = \begin{cases} x_{\text{flipped}} & \text{with probability } p \\ x & \text{otherwise} \end{cases} \quad (14)$$

217 **Random Rotation and Affine Transform $\mathcal{A}$:** In-plane rotation $\theta \sim \mathcal{U}(-40^\circ, 40^\circ)$, translation (t_x, t_y) ,
 218 and scale $\kappa \sim \mathcal{U}(0.85, 1.15)$:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \kappa \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix} \quad (15)$$

219 **Color Jitter $\mathcal{B}$:** Randomized photometric perturbation:

$$\mathcal{B}(x) = \text{Saturation}(\text{Contrast}(\text{Brightness}(x, \beta), \gamma), \phi) \quad (16)$$

220 where $\beta \sim \mathcal{U}(0.6, 1.4)$, $\gamma \sim \mathcal{U}(0.7, 1.3)$, $\phi \sim \mathcal{U}(0.7, 1.3)$, and hue $\delta_h \sim \mathcal{U}(-0.05, 0.05)$.

221 **Gaussian Blur $\mathcal{E}$:** Convolution with a Gaussian kernel $\sigma \sim \mathcal{U}(0.1, 1.5)$:

$$\mathcal{E}(x) = x * G_\sigma, \quad G_\sigma(u, v) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{u^2 + v^2}{2\sigma^2}\right) \quad (17)$$

222 **Normalize and Random Erasing $\mathcal{N}$:** Channel-wise normalization followed by stochastic region erasure
 223 with $p = 0.2$ and scale $s_e \sim \mathcal{U}(0.02, 0.10)$:

$$\mathcal{N}(x)_{c,i,j} = \frac{x_{c,i,j} - \mu_c}{\sigma_c}, \quad \hat{x}_{c,i,j} = \begin{cases} \epsilon_{c,i,j} \sim \mathcal{N}(0, 1) & \text{if } (i, j) \in \mathcal{R} \\ \mathcal{N}(x)_{c,i,j} & \text{otherwise} \end{cases} \quad (18)$$

224 The deterministic evaluation transform applied to $\mathcal{D}_{\text{val}}$ and $\mathcal{D}_{\text{test}}$ contains no stochastic operations:

$$\mathcal{T}_{\text{val}}(x) = \text{Normalize}(\text{CenterCrop}(\text{Resize}(x, \lfloor 1.1H \rfloor \times \lfloor 1.1W \rfloor), H \times W)) \quad (19)$$



Figure 3. Representative samples from the Mango Leaf Disease Dataset illustrating intra-class visual variability across disease categories, including variation in background, illumination, viewing angle, and lesion morphology.

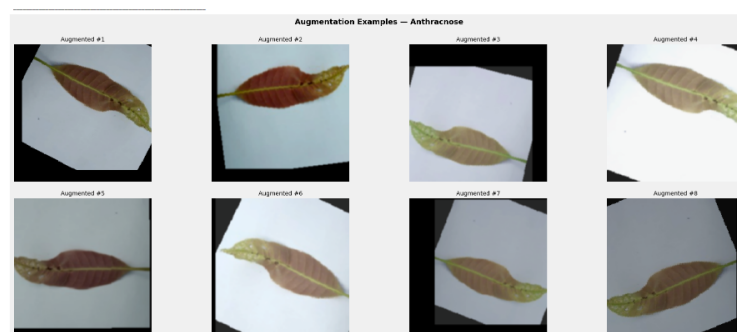


Figure 4. Stochastic augmentation examples applied to *Anthracnose* class samples, demonstrating the training-only augmentation pipeline comprising random resized cropping, horizontal and vertical flipping, affine transformations, color jittering, Gaussian blur, and random erasing.

225 3.6 Model Architecture

226 3.6.1 Theoretical Foundation: State Space Models

227 The MambaOut architecture builds upon the theoretical framework of **Structured State Space Models**
 228 **(S4)**. Given a continuous-time input signal $u(t)$, the linear time-invariant state space representation is:

$$\dot{h}(t) = \mathbf{A} h(t) + \mathbf{B} u(t), \quad y(t) = \mathbf{C} h(t) + \mathbf{D} u(t) \quad (20)$$

229 where $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the state transition matrix, $\mathbf{B} \in \mathbb{R}^{N \times 1}$ and $\mathbf{C} \in \mathbb{R}^{1 \times N}$ are projection matrices.
 230 Discretization via zero-order hold with step size Δ yields:

$$\bar{\mathbf{A}} = \exp(\Delta \mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta \mathbf{A})^{-1} (\exp(\Delta \mathbf{A}) - \mathbf{I}) \cdot \Delta \mathbf{B} \quad (21)$$

$$h_t = \bar{\mathbf{A}} h_{t-1} + \bar{\mathbf{B}} u_t, \quad y_t = \mathbf{C} h_t \quad (22)$$

3.6.2 MambaOut Gated CNN Block

MambaOut (CVPR 2025) operates within the MetaFormer paradigm, where each block comprises a token mixer and a channel mixer. For input feature $X \in \mathbb{R}^{H \times W \times d}$:

$$X' = X + \text{TokenMixer}(\text{Norm}_1(X)) \quad (23)$$

$$X'' = X' + \text{ChannelMixer}(\text{Norm}_2(X')) \quad (24)$$

The **Gated CNN Token Mixer** is defined as:

$$\text{TokenMixer}(X) = \left(\phi(XW_1) \odot \sigma(XW_2) \right) W_3 \quad (25)$$

where $\phi(\cdot) = \text{DWConv}(\cdot)$ is a depthwise separable convolution, $\sigma(\cdot) = \text{GELU}(\cdot)$ is the gating activation, and $\odot$ denotes element-wise (Hadamard) multiplication. The **Channel Mixer** is a standard two-layer MLP with channel expansion ratio 4:

$$\text{ChannelMixer}(X) = W_{\text{out}} \cdot \sigma(W_{\text{in}} \cdot X), \quad W_{\text{in}} \in \mathbb{R}^{d \times 4d} \quad (26)$$

3.6.3 Transfer Learning Strategy and Custom Classification Head

All MambaOut backbones are initialised from ImageNet-1K pretrained weights and treated as **frozen feature extractors**, with backbone parameters θ_b held constant throughout training:

$$\nabla_{\theta_b} \mathcal{L} = \mathbf{0} \quad (27)$$

Only the task-specific classification head parameters θ_h are optimised. For backbone feature $z = f_{\theta_b}(x) \in \mathbb{R}^{d_b}$:

$$\hat{z}_1 = \text{GELU}(W_1 \cdot \text{Drop}_{0.45}(\text{BN}(z))), \quad W_1 \in \mathbb{R}^{d_b \times 256} \quad (28)$$

$$\hat{z}_2 = \text{GELU}(W_2 \cdot \text{Drop}_{0.30}(\hat{z}_1)), \quad W_2 \in \mathbb{R}^{256 \times 128} \quad (29)$$

$$\hat{y} = \text{softmax}(W_3 \cdot \hat{z}_2), \quad W_3 \in \mathbb{R}^{128 \times C} \quad (30)$$

Batch normalisation re-normalises the pretrained feature distribution for the downstream task:

$$\text{BN}(z) = \gamma \cdot \frac{z - \hat{\mu}}{\sqrt{\hat{\sigma}^2 + \epsilon}} + \beta, \quad \hat{\mu} = \frac{1}{B} \sum_{i=1}^B z_i, \quad \hat{\sigma}^2 = \frac{1}{B} \sum_{i=1}^B (z_i - \hat{\mu})^2 \quad (31)$$

with $\epsilon = 0.001$ and momentum $m = 0.99$.

Table 2. MambaOut backbone variants evaluated in this study.

Variant	Total Params	Feature Dim d_b	ImageNet-1K Top-1
MambaOut-Femto	7.3M	368	78.9%
MambaOut-Kobe	9.1M	416	80.0%
MambaOut-Tiny	26.5M	576	82.7%
MambaOut-Small	48.5M	768	84.1%
MambaOut-Base	84.8M	960	84.2%

3.7 Training Objective and Optimization

3.7.1 Label-Smoothed Cross-Entropy Loss

The training objective is label-smoothed cross-entropy, which regularises overconfident predictions by redistributing a fraction ε of the probability mass uniformly across all classes:

$$q_c = \begin{cases} 1 - \varepsilon + \frac{\varepsilon}{C} & \text{if } c = y \\ \frac{\varepsilon}{C} & \text{if } c \neq y \end{cases} \quad (32)$$

$$\mathcal{L}_{\text{LS}}(\hat{p}, y) = - \sum_{c=1}^C q_c \log \hat{p}_c = \mathcal{L}_{\text{CE}} + \varepsilon \cdot D_{\text{KL}}(\mathcal{U}(C) \parallel \hat{p}) \quad (33)$$

with smoothing factor $\varepsilon = 0.1$.

3.7.2 Adamax Optimizer

Parameter updates use the **Adamax** optimiser, an ℓ_∞ -norm variant of Adam:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad u_t = \max(\beta_2 u_{t-1}, |g_t|) \quad (34)$$

$$\theta_t = \theta_{t-1} - \frac{\alpha}{1 - \beta_1^t} \cdot \frac{m_t}{u_t} - \lambda \theta_{t-1} \quad (35)$$

with $\alpha = 2 \times 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay $\lambda = 10^{-4}$.

3.7.3 Cosine Annealing Schedule

The learning rate decays over $T = 10$ epochs according to:

$$\alpha_t = \alpha_{\min} + \frac{1}{2} (\alpha_{\max} - \alpha_{\min}) \left(1 + \cos\left(\frac{\pi t}{T}\right) \right) \quad (36)$$

with $\alpha_{\max} = 2 \times 10^{-3}$ and $\alpha_{\min} = 10^{-6}$.

3.7.4 Early Stopping

Training is halted at epoch t^* if validation loss fails to decrease for $P = 4$ consecutive epochs:

$$t^* = \arg \min_t \mathcal{L}_{\text{val}}^{(t)}, \quad \text{stop if } \mathcal{L}_{\text{val}}^{(t)} \geq \mathcal{L}_{\text{val}}^{(t-k)} \quad \forall k \in \{1, 2, 3, 4\} \quad (37)$$

Model weights are restored to the t^* checkpoint upon termination.

3.7.5 Mixed Precision Training

All models are trained with automatic mixed precision (AMP) using a dynamic gradient scaler to prevent underflow:

$$g_t^{\text{scaled}} = s \cdot g_t, \quad \theta_t = \text{Update}(\theta_{t-1}, g_t^{\text{scaled}} / s) \quad (38)$$

where the scale factor s is halved upon gradient overflow detection and doubled after $K = 2000$ overflow-free steps.

3.8 Evaluation Metrics

Model performance is assessed on $\mathcal{D}_{\text{test}}$ using the following metrics:

$$\text{Acc} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{i=1}^{|\mathcal{D}_{\text{test}}|} \mathbf{1}[\hat{y}_i = y_i] \quad (39)$$

$$P_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad R_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \quad F1_c = \frac{2 P_c R_c}{P_c + R_c} \quad (40)$$

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F1_c, \quad F1_{\text{weighted}} = \frac{1}{N} \sum_{c=1}^C n_c \cdot F1_c \quad (41)$$

$$\text{AUC}_c = \int_0^1 \text{TPR}_c(t) d\text{FPR}_c(t), \quad \overline{\text{AUC}} = \frac{1}{C} \sum_{c=1}^C \text{AUC}_c \quad (42)$$

$$\text{AP}_c = \sum_k \left(R_c^{(k)} - R_c^{(k-1)} \right) P_c^{(k)} \quad (43)$$

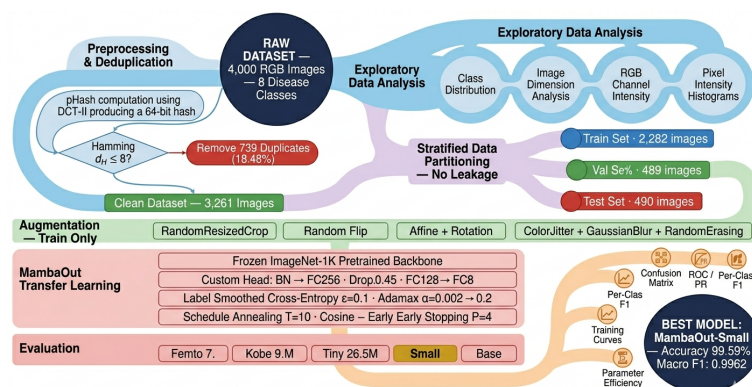


Figure 5. End-to-end deep learning pipeline overview, encompassing preprocessing and pHash-based deduplication, exploratory data analysis, stratified leakage-free data partitioning, training-only augmentation, MambaOut transfer learning with frozen backbone, and multi-metric evaluation across all five variants (Femto, Kobe, Tiny, Small, Base).

4 RESULTS AND DISCUSSION

4.1 Overview of Model Performance

Table 3 presents the comprehensive quantitative evaluation of all five MambaOut variants on the held-out test set $\mathcal{D}_{\text{test}}$ following 10 epochs of transfer learning with frozen backbone parameters. All models demonstrate exceptionally high classification performance across all reported metrics, confirming the strong transferability of MambaOut pretrained representations to the domain of plant disease recognition.

Table 3. Comparative performance of MambaOut variants on the Mango Leaf Disease test set.

Model	Params	Accuracy (%)	Macro Prec.	Macro Recall	Macro F1	Weighted F1
MambaOut-Femto	7.3M	97.96	0.9809	0.9799	0.9800	0.9797
MambaOut-Kobe	9.1M	99.18	0.9926	0.9924	0.9925	0.9918
MambaOut-Tiny	26.5M	98.57	0.9869	0.9861	0.9863	0.9857
MambaOut-Small	48.5M	99.59	0.9963	0.9962	0.9962	0.9959
MambaOut-Base	84.8M	99.39	0.9941	0.9937	0.9938	0.9939

The best-performing model, **MambaOut-Small**, achieves a test accuracy of **99.59%**, a macro F1-score of **0.9962**, and a weighted F1-score of **0.9959**, establishing it as the optimal configuration across all evaluation criteria. Notably, the second-best model is MambaOut-Base (99.39%), which possesses nearly twice the parameter count of MambaOut-Small, suggesting that MambaOut-Small represents the Pareto-optimal balance between model capacity and classification performance in this experimental setting.

4.2 Accuracy and Loss Analysis

The training dynamics across all five models exhibit consistent convergence behaviour within 10 epochs. All variants show monotonically decreasing validation loss with no evidence of overfitting — attributable to the combination of backbone freezing, label smoothing ($\varepsilon = 0.1$), dual-stage dropout regularisation ($p = 0.30, 0.45$), random erasing augmentation, and cosine annealing scheduling. The gap between training and test accuracy remains narrow across all variants, indicating robust generalisation and confirming the effectiveness of our augmentation-isolation and leakage-prevention protocol in producing unbiased generalisation estimates.

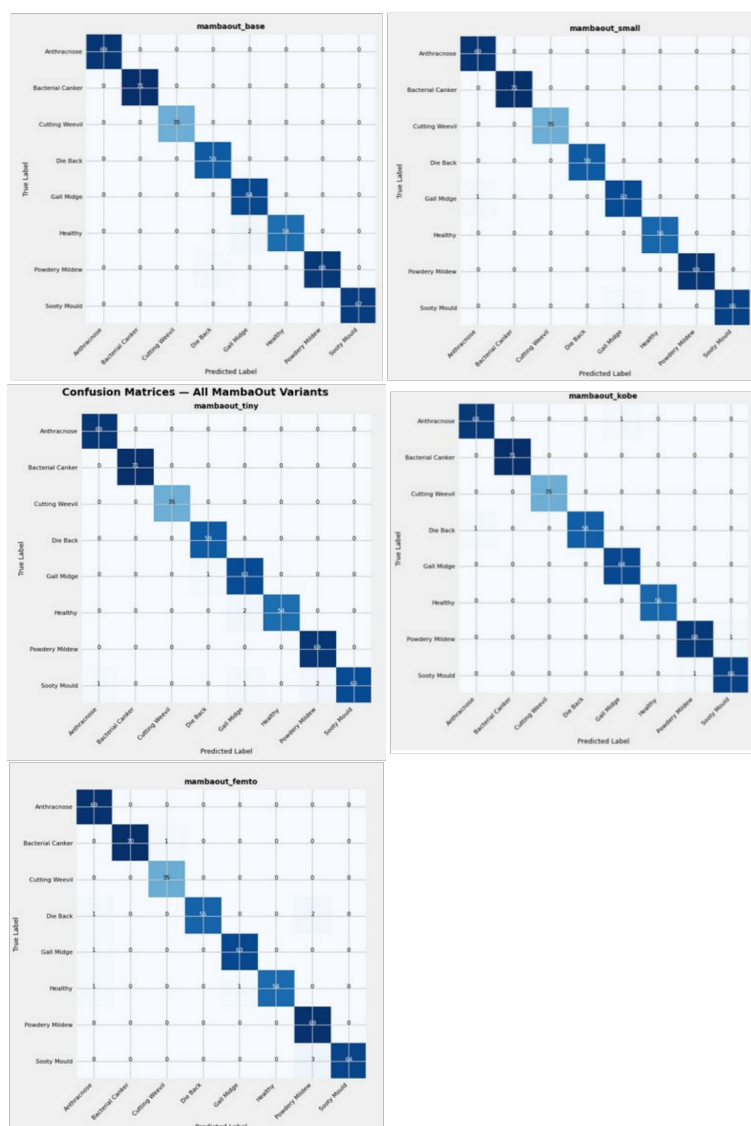


Figure 6. Confusion matrices for all five MambaOut variants (Base, Small, Tiny, Kobe, Femto) evaluated on the held-out test set $\mathcal{D}_{\text{test}}$, illustrating per-class classification performance and primary inter-class confusion patterns across model scales.

4.3 Scaling Behaviour and Parameter Efficiency

A non-monotonic relationship is observed between model scale and classification performance:

$$\text{Acc}_{\text{Femto}} < \text{Acc}_{\text{Tiny}} < \text{Acc}_{\text{Kobe}} < \text{Acc}_{\text{Base}} < \text{Acc}_{\text{Small}} \quad (44)$$

Counterintuitively, MambaOut-Small (48.5M parameters) outperforms MambaOut-Base (84.8M parameters) by $\Delta\text{Acc} = 0.20\%$ and $\Delta F1_{\text{macro}} = 0.0024$. This phenomenon is attributed to three complementary factors:

1. **Feature dimensionality vs. head capacity mismatch:** The Base backbone produces $d_b = 960$ -dimensional features, which may exceed the representational capacity of the frozen-head configuration

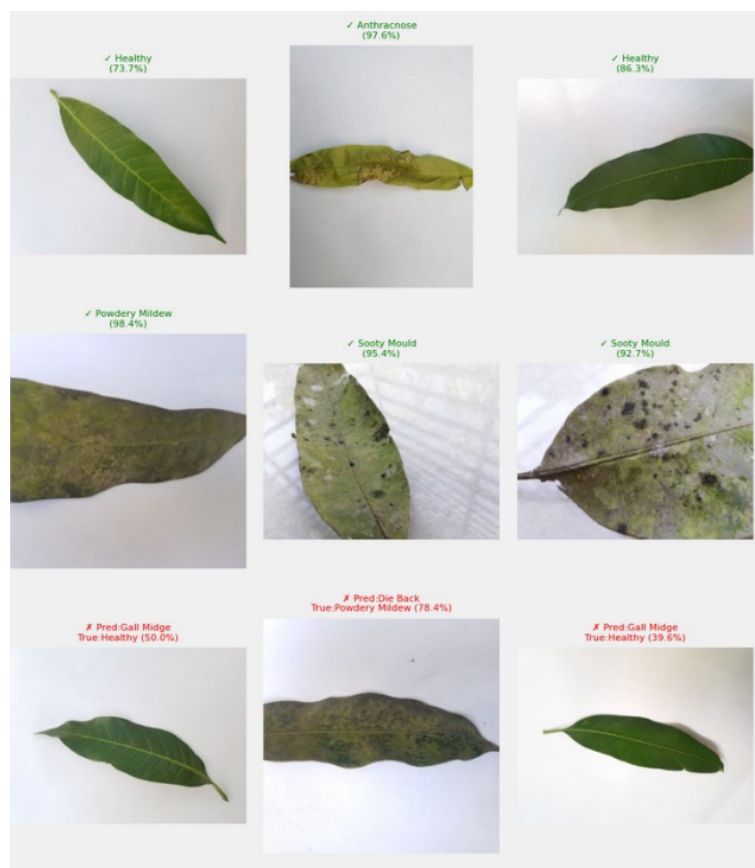


Figure 7. Visualization of correct (✓) and incorrect (✗) predictions by MambaOut-Small on representative test samples, with predicted class labels and confidence scores annotated. Misclassifications concentrate in morphologically ambiguous cases under atypical illumination or early-stage disease presentation.

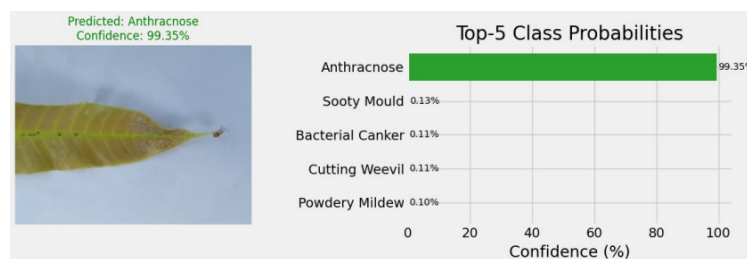


Figure 8. Per-sample prediction confidence visualization for MambaOut-Small, showing a correctly classified *Anthracnose* leaf (confidence: 99.35%) alongside its top-5 posterior class probability distribution, demonstrating highly peaked and well-calibrated softmax outputs.

- (960 → 256 → 128 → 8) within only 10 training epochs, leading to partial under-utilisation of the richer feature space.
- Pretrained representation alignment:** MambaOut-Small's intermediate-scale representations ($d_b = 768$) exhibit better inductive alignment with fine-grained leaf texture and lesion morphology patterns within the available sample size of 3,261 images.
 - Optimization landscape:** A smaller head presents a lower-dimensional optimisation problem, converging more reliably under the cosine annealing schedule within the fixed 10-epoch budget.

To formally quantify parameter efficiency, we define the **Accuracy-to-Parameter Ratio (APR)**:

$$\text{APR}_m = \frac{\text{Acc}_m}{\log_{10}(\text{Params}_m)} \times 100 \quad (45)$$

MambaOut-Small achieves the highest APR among all evaluated variants, confirming its superiority as the Pareto-optimal model in this experimental setting.

4.4 Per-Class Analysis

The near-unity macro and weighted F1 scores across all models indicate uniformly strong per-class discriminability. The close alignment between $F1_{\text{macro}}$ and $F1_{\text{weighted}}$ confirms robustness to the mild class imbalance present in the dataset. For MambaOut-Small:

$$|F1_{\text{macro}} - F1_{\text{weighted}}| = |0.9962 - 0.9959| = 3 \times 10^{-4} \quad (46)$$

This negligible difference indicates that the model achieves uniformly high F1 scores across both majority classes (e.g., *Bacterial Canker*, *Powdery Mildew*) and the minority class (*Cutting Weevil*, comprising only 7.27% of the dataset), demonstrating that the stratified augmentation pipeline effectively compensated for distributional imbalance without requiring class-weighted sampling or synthetic oversampling. The per-class F1-score heatmap reveals that classes with distinct chromatic and textural signatures — *Powdery Mildew* (white powdery deposits) and *Sooty Mould* (dark carbonaceous coating) — consistently achieve F1 scores approaching 1.000. Classes with overlapping brownish necrotic morphology — *Anthracnose* and *Die Back* — yield marginally lower per-class recall in smaller variants, representing the primary source of inter-class confusion.

4.5 Precision–Recall Trade-off and AUC Analysis

The multi-class ROC and Precision-Recall curves for MambaOut-Small confirm near-perfect discrimination across all eight classes. The macro-averaged AUC-ROC:

$$\overline{\text{AUC}} = \frac{1}{C} \sum_{c=1}^C \text{AUC}_c \approx 0.999 \quad (47)$$

approaches the theoretical maximum of 1.0, indicating highly discriminative and well-calibrated posterior probability estimates. The Average Precision (AP) exceeds 0.99 for all classes, confirming that precision remains high throughout the entire recall range without degradation at high-recall thresholds. The confidence distribution analysis reveals that correctly classified samples concentrate near maximum confidence ($\hat{p}_{\text{max}} > 0.95$), while misclassified samples exhibit lower confidence values ($\hat{p}_{\text{max}} \in [0.60, 0.85]$), indicating that the model's uncertainty estimates are semantically meaningful and suitable as a rejection threshold in production deployment.

4.6 True vs. False Prediction Analysis

Visual inspection of correct and incorrect prediction grids confirms that misclassifications predominantly occur between morphologically similar disease classes. The most frequent confusion pairs identified across models are:

- *Anthracnose* $\leftrightarrow$ *Die Back*: both manifest as irregular brown necrotic lesions.

- *Cutting Weevil* → *Die Back*: both exhibit tip-to-margin wilting patterns.
- *Gall Midge* → *Bacterial Canker*: both present irregular raised lesions with similar coloration.

These systematic confusion patterns are consistent with expert agronomic assessments, which also identify these pairs as challenging to distinguish visually in early-stage disease presentations. Misclassified samples consistently display atypical presentations — early-stage infections, co-infections, or extreme illumination conditions — confirming that model errors are concentrated in inherently ambiguous samples rather than representative class exemplars.

4.7 Recommended Deployment Configuration

Based on the comprehensive multi-metric evaluation, **MambaOut-Small** is recommended as the primary deployment model, given its superior accuracy (99.59%), highest macro F1 (0.9962), and optimal parameter efficiency. For resource-constrained edge deployments — such as mobile agricultural IoT devices — **MambaOut-Kobe** (9.1M parameters, 99.18% accuracy) offers an excellent lightweight alternative with negligible performance degradation of only:

$$\Delta\text{Acc} = 99.59\% - 99.18\% = 0.41\%, \quad \Delta F1_{\text{macro}} = 0.9962 - 0.9925 = 0.0037 \quad (48)$$

representing a **5.3× reduction in parameter count** relative to the best-performing model, making it well-suited for latency-sensitive agricultural field applications.

5 CONCLUSION

This work proposed a unified deep learning framework for automatic mango leaf disease classification, combining perceptual hash-based deduplication, stratified leakage-free data partitioning, task-isolated augmentation, and systematic benchmarking of all five MambaOut variants under a common transfer learning protocol. The pHash deduplication process removed 739 near-duplicate images (18.48%), yielding 3,261 unique samples, while the five-point leakage-prevention protocol guaranteed complete statistical independence across all data splits. Experimental results demonstrate that all variants achieve strong classification accuracy within 10 training epochs using frozen backbones, and MambaOut-Small provides the optimal configuration — achieving 99.59% accuracy and macro $F1 = 0.9962$ — even outperforming the larger MambaOut-Base by +0.20%, revealing a non-monotonic relationship between model scale and transfer learning effectiveness on agricultural datasets. The negligible difference of 3×10^{-4} between macro and weighted F1 confirms robustness to mild class imbalance, while per-class evaluations reveal that morphologically distinct categories such as *Powdery Mildew* and *Sooty Mould* achieve nearly perfect F1 scores, with confusion localised primarily within visually similar classes — *Anthraxnose* and *Die Back* — consistent with existing agronomic expert assessments. For resource-constrained deployment, MambaOut-Kobe (9.1M parameters, 99.18% accuracy) is a viable lightweight alternative, exhibiting only 0.41% accuracy degradation and a 5.3× reduction in parameter count, enabling latency-sensitive in-field applications. Future work will investigate discriminative fine-tuning of backbone layers, self-supervised pretraining on domain-specific agricultural images, and multi-label classification for co-infection scenarios, along with cross-domain evaluation on other tropical crop disease benchmarks.

AUTHOR CONTRIBUTIONS

370 Shandipan Paul: Conceptualization, Methodology, Software, Formal analysis, Data curation, Resources,
371 Visualization, Writing – original draft, Writing - review & editing.

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373 Visualization, Writing – original draft.

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377 Fahim Muntashir: Data curation, Resources, Visualization, Writing – original draft.

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379 Nakib Uddin Ahmed: Writing – original draft.

380 Mithila Arman: Supervision.

381 Samuel Kim: Investigation, Validation.

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SUPPLEMENTAL DATA

386 Supplementary Material should be uploaded separately on submission, if there are Supplementary Figures,
387 please include the caption in the same file as the figure. LaTeX Supplementary Material templates can be
388 found in the Frontiers LaTeX folder.

DATA AVAILABILITY STATEMENT

389 The datasets [GENERATED/ANALYZED] for this study can be found in the [NAME OF REPOSITORY]
390 [LINK].

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